Introduction

Doubly Linked List
Data Structures and its Applications (UE25CS52A)

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Quiz : Doubly Linked List

1. In a non-circular doubly linked list, the prev pointer of the first node contains:

a) Pointer to the last node

b) Pointer to the second node

c) NULL

d) Pointer to itself

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Quiz : Doubly Linked List

2. Suppose p is a pointer to a node in the middle of the list and we want to insert new node pointed to by temp immediately after p. Which sequence is right with respect to p?

a) p->next_ = temp->next_; p->prev_ = temp->prev_;

b) temp->next_ = p->next_; temp->prev_ = p->prev_;

c) temp->next_ = p->next_; temp->prev_ = p;

d) temp->next_ = p; temp->prev_ = p->prev_;

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Quiz : Doubly Linked List

3. When a new node is added to an empty list,

a) neither head_ not tail_ of the DLL will change

b) both head_ and tail_ of the DLL could change

c) only head_ of the DLL would change

d) Only tail_ of the DLL would change

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Quiz : Doubly Linked List

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Quiz : Doubly Linked List

4. temp->next_ = p->next_; temp->prev_ = p; p->next_ = temp;

While inserting a new node temp not at the end of the list, which statement is missing?

a) p->next_->prev_ = temp;

b) p->prev_ = temp;

c) p->next_ = NULL;

d) p->next_ = NULL;

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Quiz : Doubly Linked List

4. temp->next = p->next; temp->prev = p; p->next = temp;

While inserting a new node temp not at the end of the list, which statement is missing?

a) p->next_->prev_ = temp;

b) p->prev_ = temp;

c) p->next_ = NULL;

d) p->next_ = NULL;

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Quiz : Doubly Linked List

5. p->prev->next = p->next;p->next->prev = p->prev;

a) Deletes the node to which p points to and handles memory correctly

b) Deletes the node to which p points to, but results in memory leak

c) Breaks the list and part of the list is lost

d) Deletes the node in the beginning, or at the end or the middle

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Quiz : Doubly Linked List

5. p->prev->next = p->next;p->next->prev = p->prev;

a) Deletes the node to which p points to and handles memory correctly

b) Deletes the node to which p points to, but results in memory leak

c) Breaks the list and part of the list is lost

d) Deletes the node in the beginning, or at the end or the middle

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Doubly Linked List : a few points

- Each node allows movement to the previous previous and the nodes
  - Normally called prev_ and next_ standing for previous and next
- The list node has two pointers – one to the first node and one to the last node
  - Normally called head_ and tail_
- Allows for traversal in both the directions forward and backward

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Doubly Linked List : interface

- Initialize / Deinitialize the list
- Insert anywhere in the list
- Delete anywhere in the list
- Traverse forward or backward
- Search for a designated key

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Doubly Linked List : key structure

```
struct node {
   int key_;
   struct node *prev_;
   struct node *next_;
};
typedef struct node node_t;
struct DLL {
            node_t *head_;
            node_t *tail_;
};
typedef struct DLL dll_t;
```


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Doubly Linked List : implementation

- Initialize / Deinitialize the list
  - Initialize : Ground both head_ and tail_
  - Deinitialize : traverse and free the nodes. Reinitialize the list

- Insert anywhere in the list
  - Take care of the following cases
    - Empty list : both head_ and tail_ should change
    - Add in the beginning : head_ should change
    - Add at the end : tail_ should change
    - Add in the middle
  - Observe there will be 4 changes to the pointers

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Doubly Linked List : implementation

- Delete anywhere in the list
  - In the beginning : head_ should change
  - At the end : tail_ should change
  - Anywhere in the middle
  - Deleting the only node : both head_ and tail_ should change
- Traverse forward or backward
- Search for a designated key

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Singly vs. Doubly Linked Lists

Worst-Case Operation (n nodes)	Singly Linked List	Doubly Linked List
Move Forward	1 step	1 step
Move Backward	n steps (Requires re-traversal)	1 step (Instant fallback)
Insert Before a Node	n steps (Find predecessor)	1 step (Using prev_ link)
Delete a Designated Node	n steps (Find predecessor)	1 step (Local isolation)
Searching for a Key	n steps	n steps

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Doubly Linked List : Advantages

1. Bidirectional traversal
2. Easy deletion
3. Easy insertion before or after a known node
4. Natural representation of navigation

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Doubly Linked List : Disadvantages

1. Extra memory

2. More pointer manipulation

3. More opportunities for bugs

4. Still has linear search

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Doubly Linked List : Applications

1. Browser History

2. Music Player Playlist

3. Undo and Redo

4. LRU cache

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THANK YOU

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